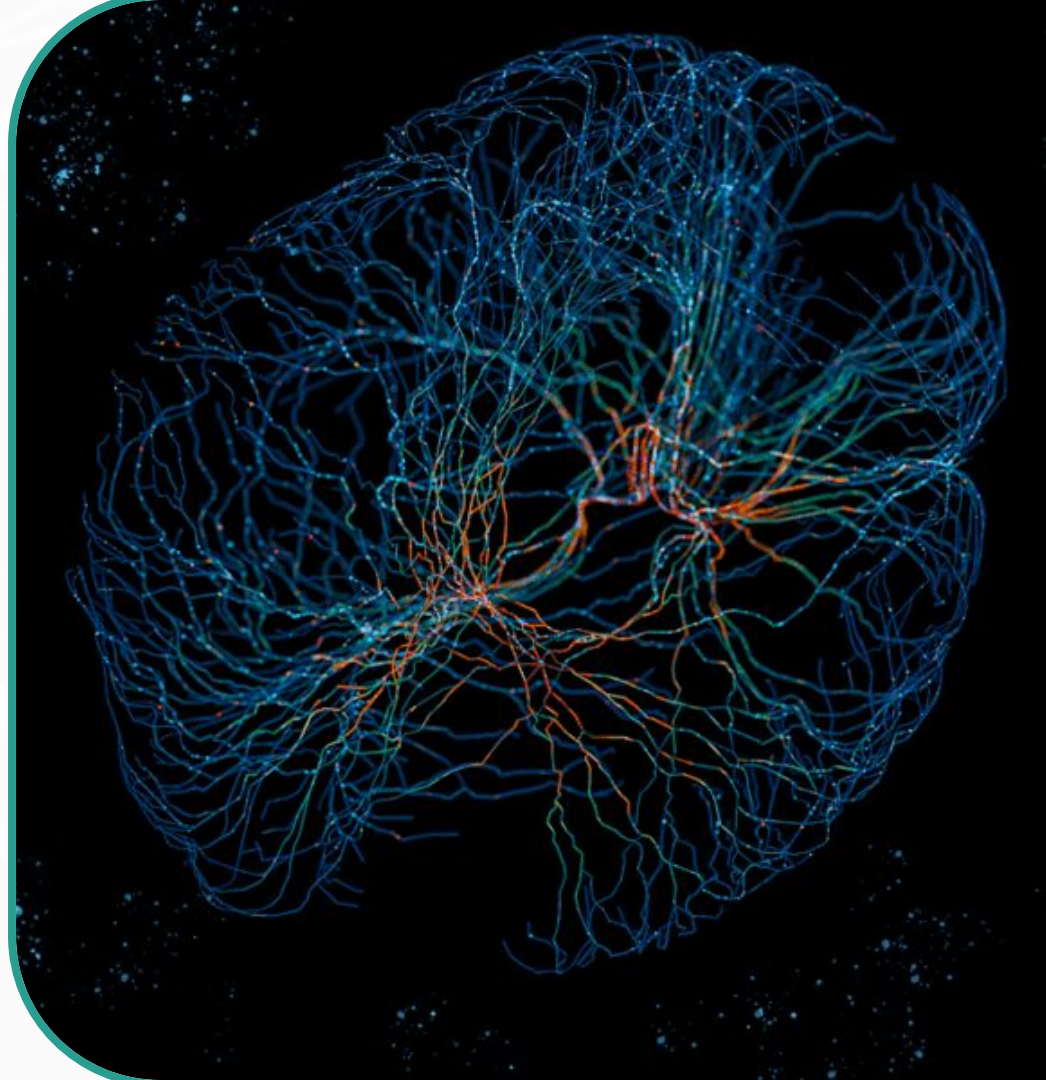




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# Neural Networks & FNN

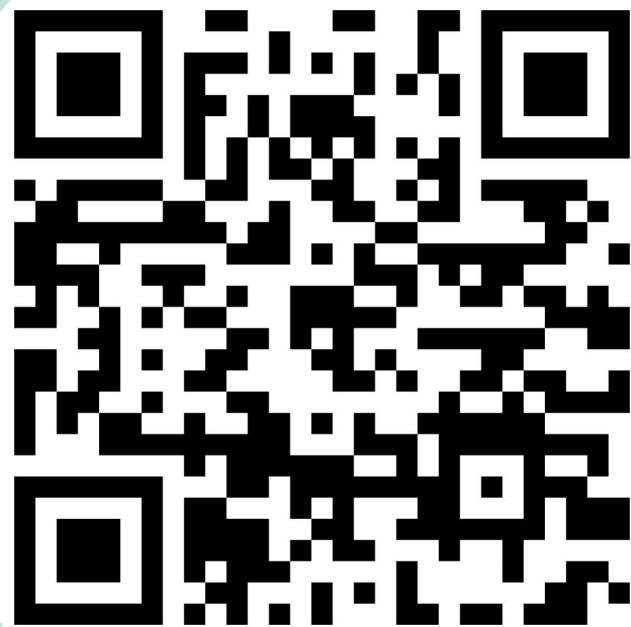


# ATTENDANC

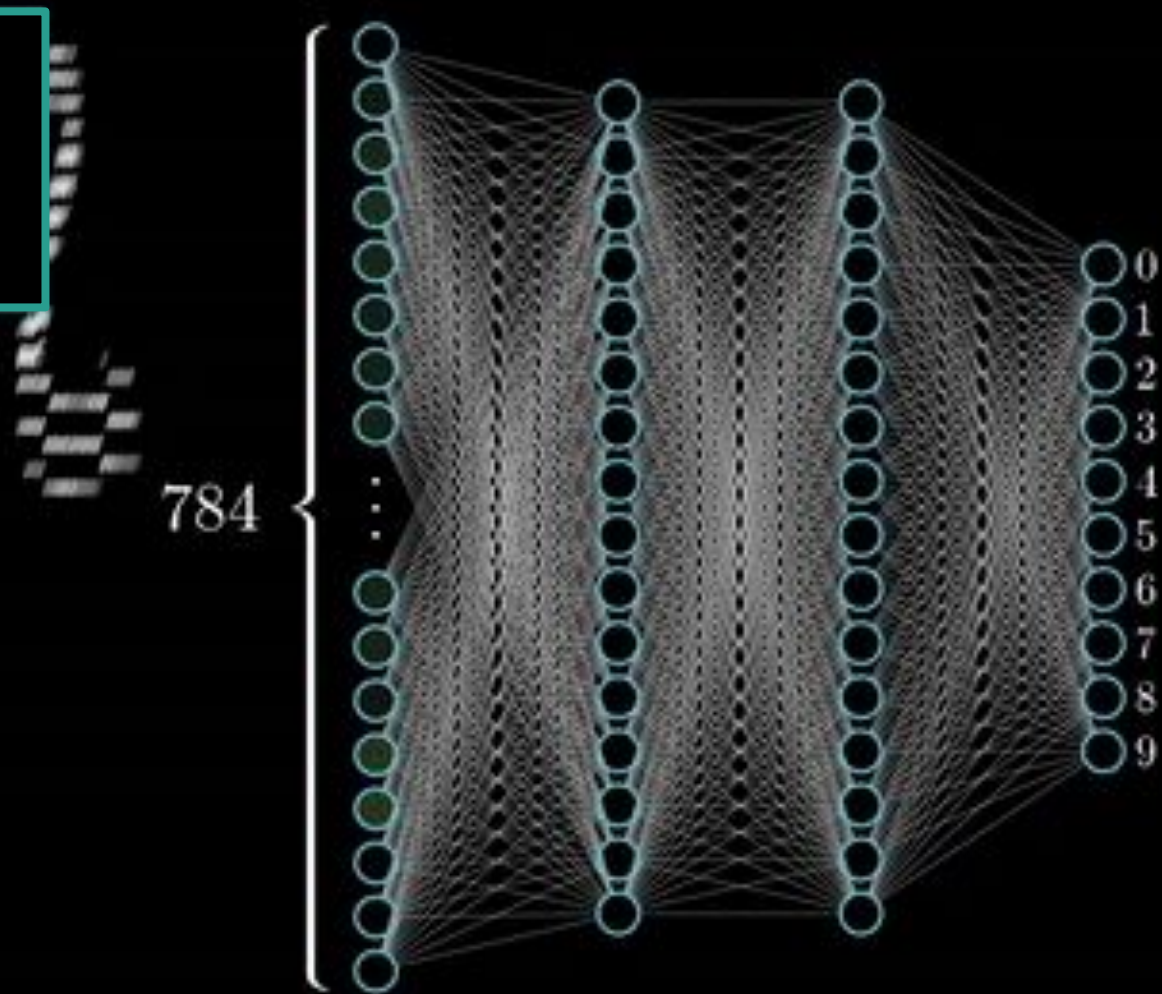
Meeting Name: "Neural Networks"



# Spring trip with NDL!



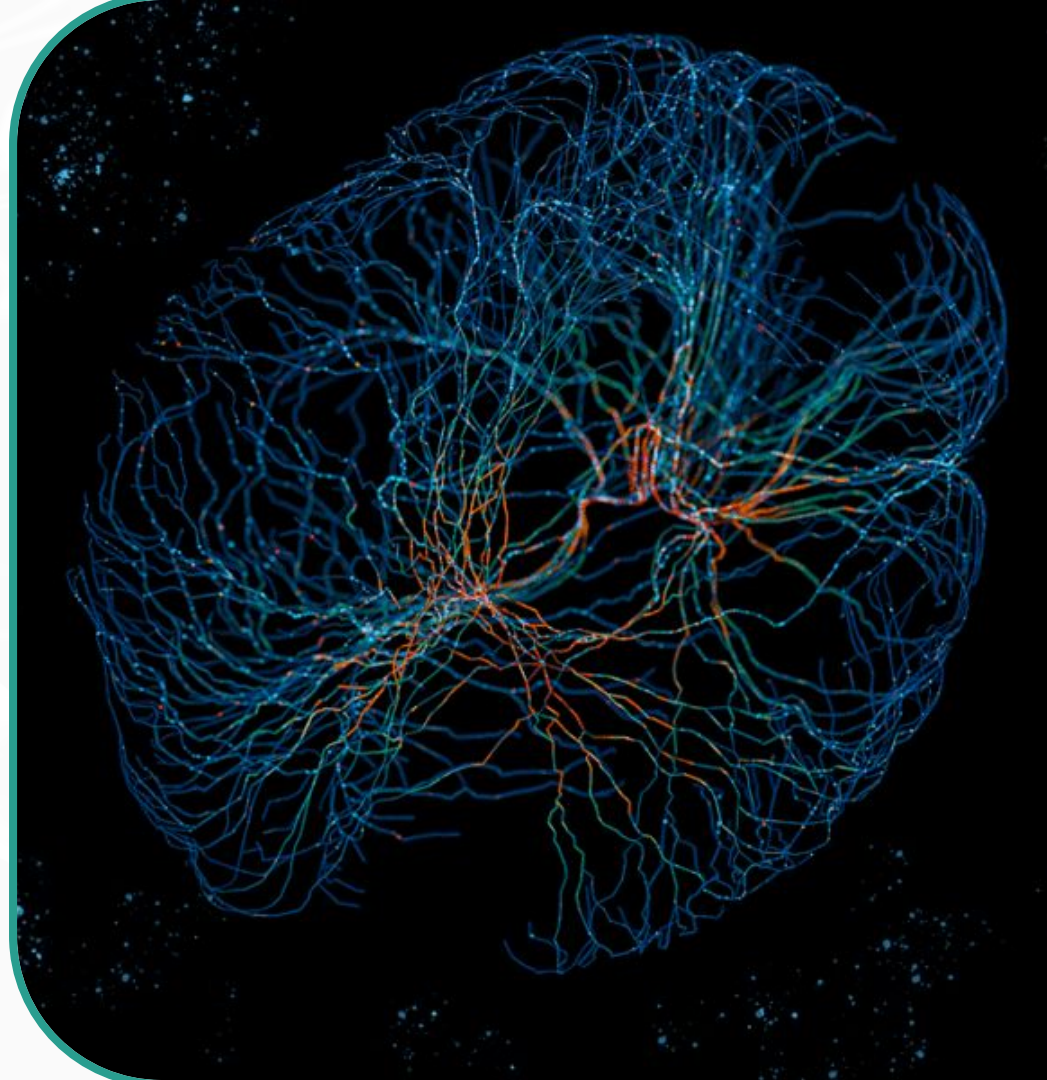
2





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# Neural Networks & FNN



# Neural networks power modern AI systems



Image Recognition



Chatbots



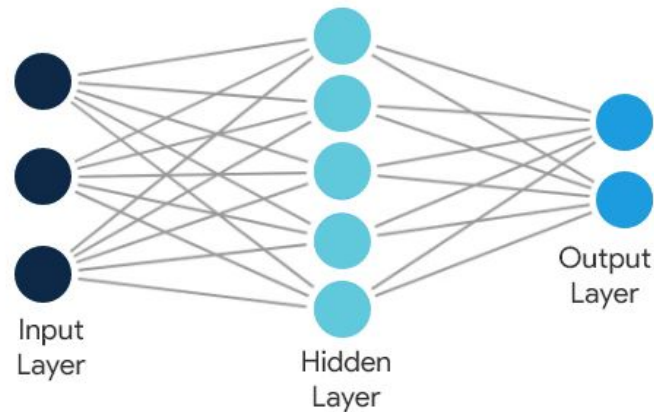
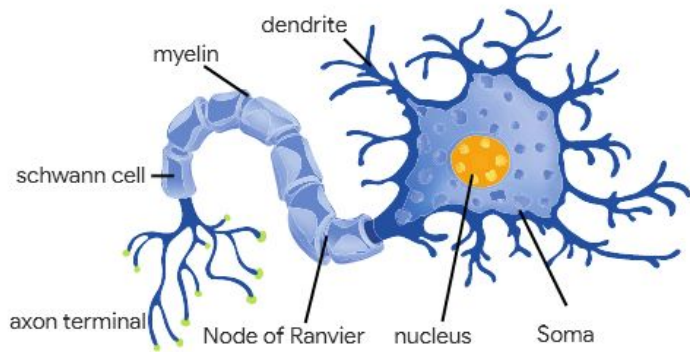
Autonomous Vehicles

# A neural network is a machine learning model inspired by how the human brain processes information

Biological Neuron

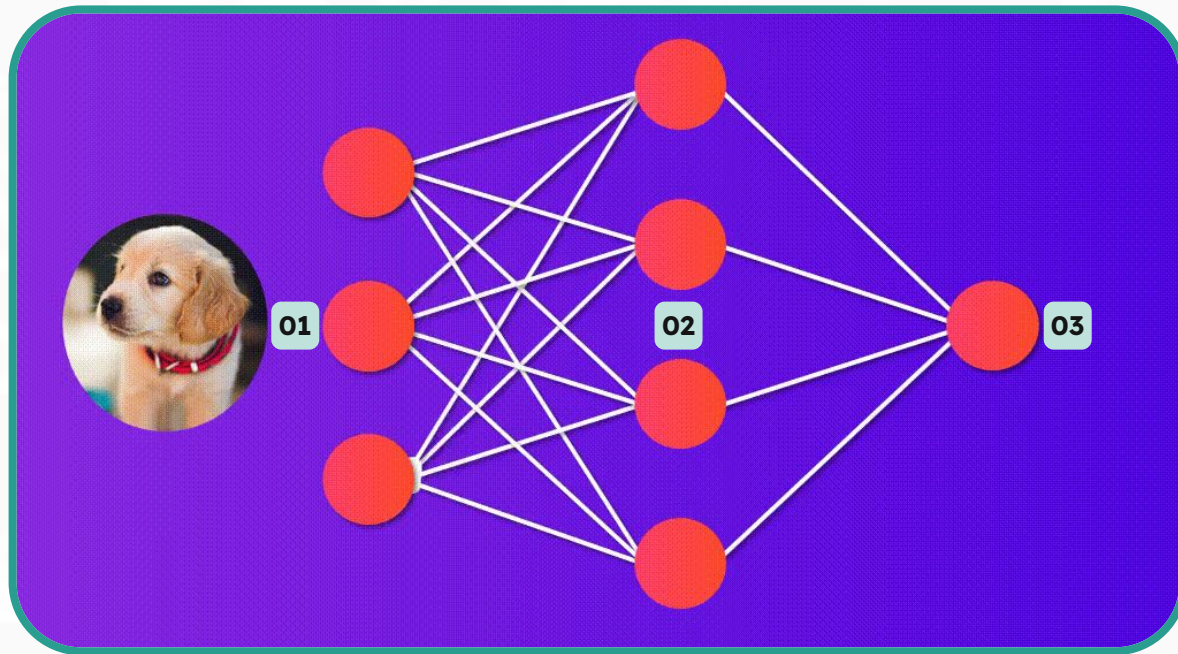
VS

Artificial Neural Network



It's purpose is to learn patterns in complex data and transform inputs into useful predictions

# Its architecture is organized into layers of neurons



## 01 Input Layer

Receives the raw data  
Each input neuron represents one feature

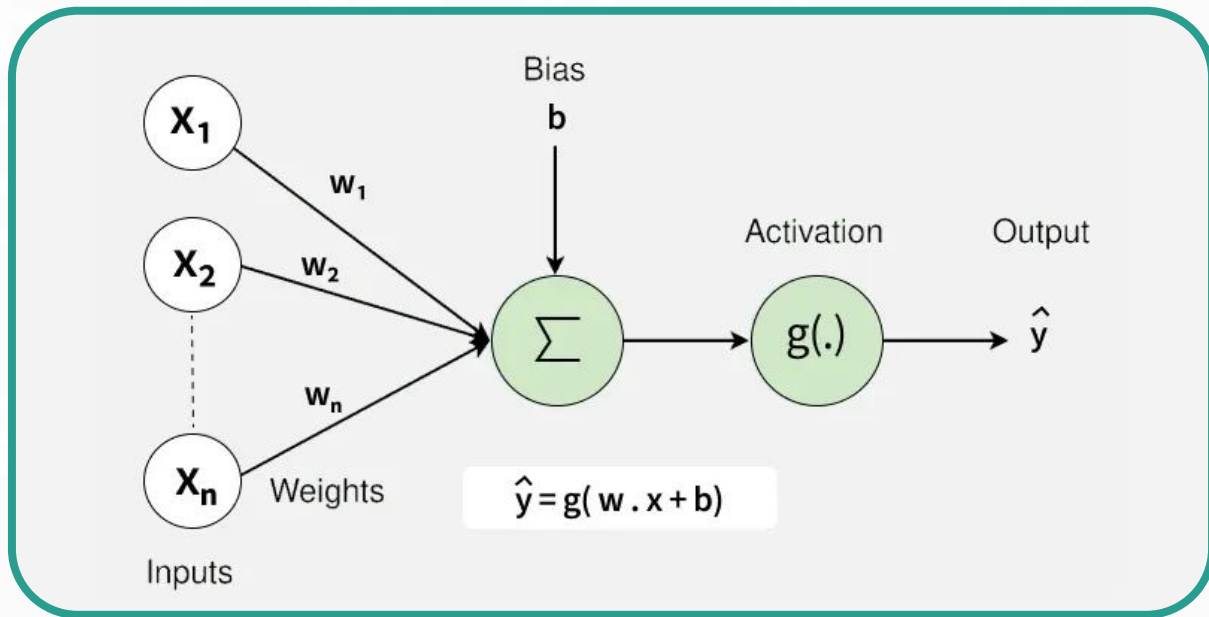
## 02 Hidden Layers

Performs intermediate calculations by  
combining inputs, detecting patterns, and  
transforming the data

## 03 Output Layer

Produces the final prediction

# A neuron (node) receives inputs, multiplies them by weights, adds a bias, and passes the result through an activation function



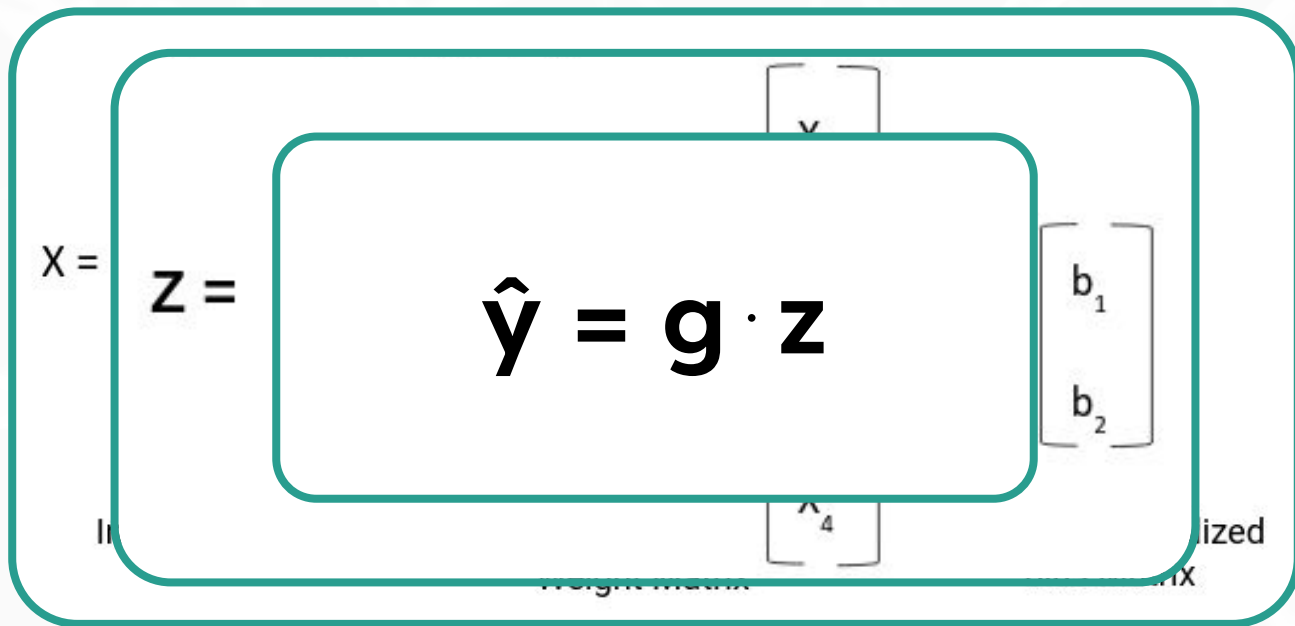
## Step 1: Weighted Sum

$x$  = input values |  $w$  = weights |  $b$  = bias

## Step 2: Activation Function

Activation functions decide how strongly the neuron activates

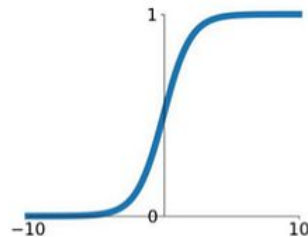
A neuron (node) receives inputs, multiplies them by weights, adds a bias, and passes the result through an activation function



# Activation function introduces non-linearity, which allows neural networks to learn complex patterns

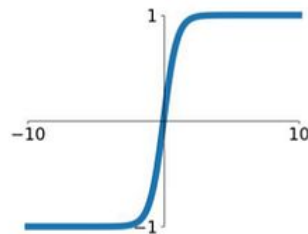
**Sigmoid**

$$\sigma(x) = \frac{1}{1+e^{-x}}$$



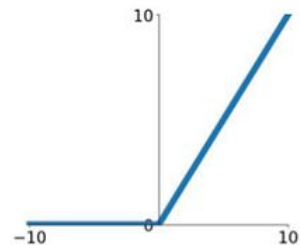
**tanh**

$$\tanh(x)$$

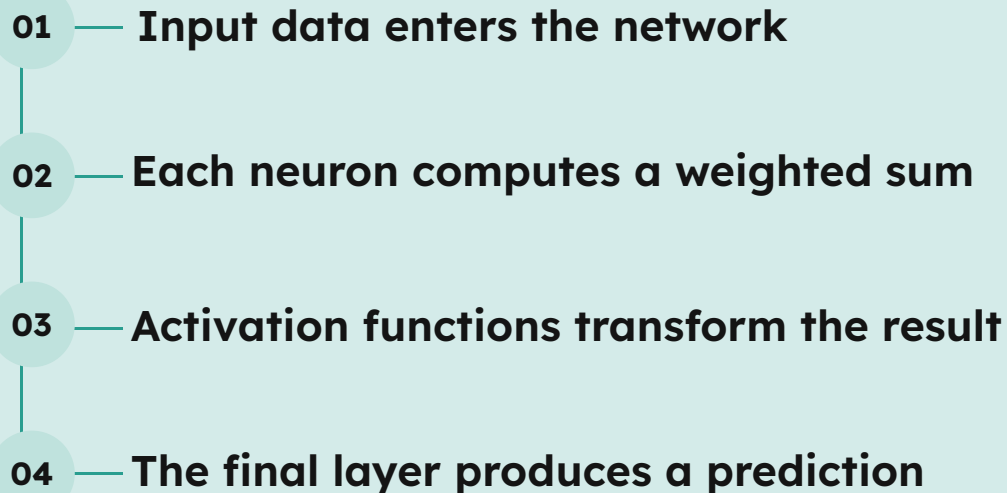


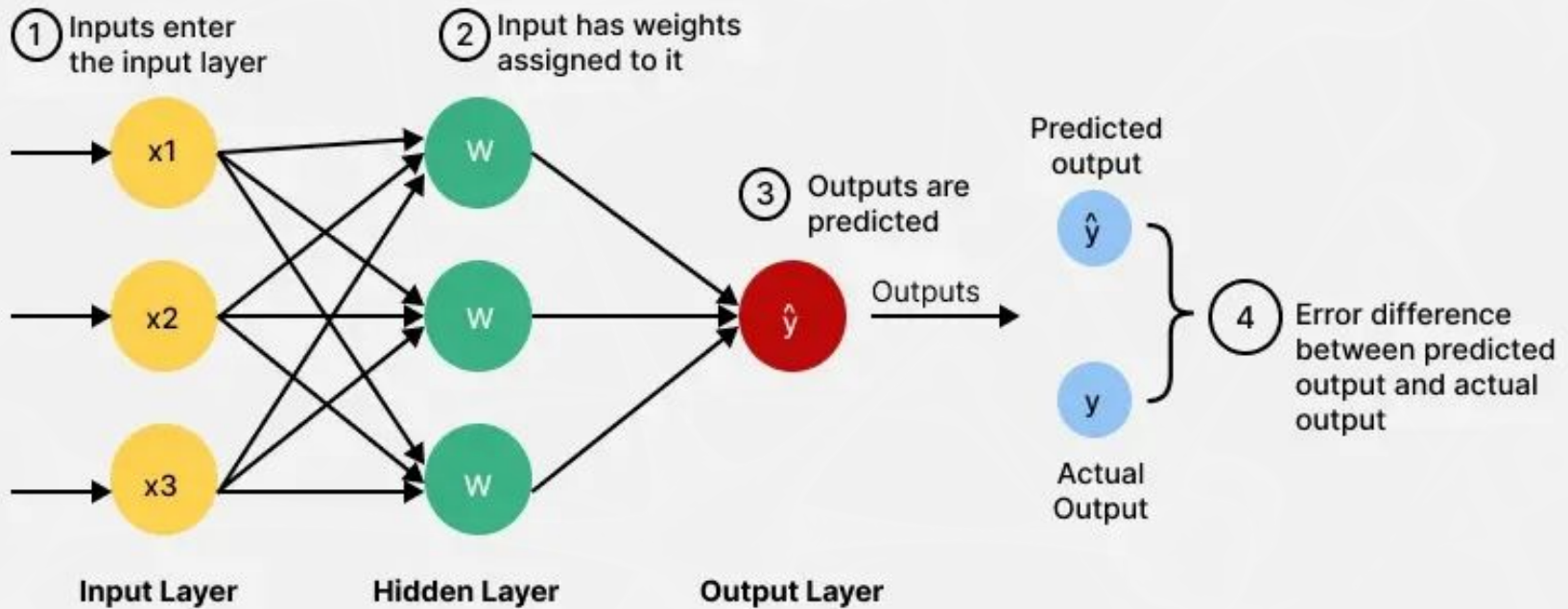
**ReLU**

$$\max(0, x)$$



# **In feedforward neural networks data flows in one direction only (no loops or cycles)**

- 
- 01 — Input data enters the network**
  - 02 — Each neuron computes a weighted sum**
  - 03 — Activation functions transform the result**
  - 04 — The final layer produces a prediction**



# Feedforward Neural Networks

## Pros

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- **Learn nonlinear relationships**
- **Handle large datasets**
- **Approximate extremely complex functions**

## Cons

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- **Overfitting:** may memorize training data rather than learning real patterns
- **Interpretability:** “black box” models
- **Data Requirements:** usually require large datasets

# Key Takeaways

## Neural Networks

- Inspired by the brain
- Learn by adjusting weights between neurons
- Use activation functions to model nonlinear relationships
- Becomes powerful when many layers are combined

## Feedforward Foundation

- The simplest neural network architecture
- The foundation of most modern deep learning systems

Come back next week when we cover convolution neural network (CNN) as well as transformer network



# Colab

Link is in the GroupMe as well



# **Thank You**

**SPRING TRIP  
APPLICATION**



**FOLLOW OUR  
INSTAGRAM**

